

where no man can go and no man wants to go. What makes DuctBot one-of-its-kind product in India is that the other alternatives available are bulky, very difficult to carry and control, and require the user to be trained prior to operating these. Our system is light and easy to carry in flight. Besides, no training is required to show how a PS2 joystick works," says Fahad Azad, managing partner, Robosoft Systems.

The technology that drives the DuctBot

"DuctBot is completely made from off-the-shelf systems available across the world. We have tried to make varied technologies work in sync with each other. It is a combination of mechanical, electronics and software engineering. The robot is designed in CAD software using embedded systems and microcontrollers and is programmed in embedded C. Our main consideration while designing the robot in CAD was whether it will be possible for us to write the code for the performance desired from the product, and will the electronics system be reliable in varied conditions?" says Azad.

"We have tried to make the device as cost-effective as possible so that

How healthy is the AC environment you are exposed to?

Azad has a tip to share: "If you go to any centrally air-conditioned environment and happen to see automated air fresheners on the walls spraying the freshener regularly, it is an indication that the air quality is bad and the content of dust is so high that there is bacteria growth in the duct, causing odour in the air.

"If there is no dust in the duct, there will never be bacteria and pollen formation. Just changing air filters regularly is not the solution to this problem; the ducts need to be thoroughly cleaned and disinfected."

the users can recover the cost of the machine in much lesser time. They can discard their old machine and go for a new one, instead of buying heavily engineered machines, which are high in maintenance and use components that become obsolete or expensive to get after some years," he adds.

The scenario and market for duct cleaners in India

Are people aware of the importance of having a duct cleaner? To this, Azad says, "The issue in India is that there is no regulation or policy of the government that makes it mandatory to check and maintain indoor air quality—mostly because air-conditioning is relatively new in India. Other countries have strict regulations and quality controls, which are essential to comply with for any establishment. Companies in India do it on their own as it reduces

electricity consumption and bad odour from environments."

To market the product, RoboSoft uses the Internet as the key medium. "Our main market is the Middle East, followed by the US, Europe and Southeast-Asian countries. Our key customers include Blue Star in India and EPSCO in the Middle East," shares Azad.

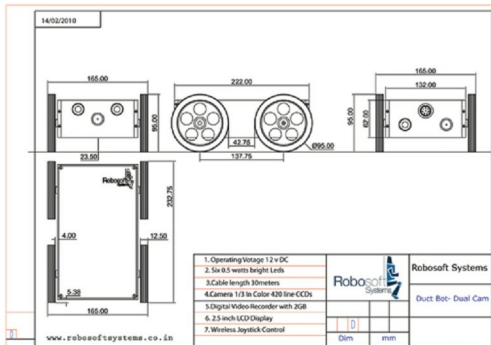
What's in the offing?

The company has not filed for any patents so far as the team feels that the international process is very expensive and the Indian patent system is not so strong.

"We intend to file for a patent soon. Right now we are bootstrapped. We believe that it is the customers who bring in the money in India and not the patents. If it were software, we could have licensed it to users," says Azad.

Azad also shares some of the other plans related to the future versions of the device: "We are planning to launch a new version of the device named DuctBot HD. It will come with an HD 1080i camera, which offers a full high-definition recording capability compared to the CCD colour camera used in the existing product.

"We have also designed a very small robot fitted with a night vision camera and microphone for National Security Guards commandos. This robot can be dropped inside any room or building before going in to preview the place. We are looking for agencies who can help us sell our products to the government." ●



DuctBot data-sheet